

Preethi Ranganathan

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October 23, 2025

Hiring Committee

University of Michigan – Robotics Department

Dear Hiring Committee,

I am writing to apply for the Software Developer position in the Robotics Department at the University of Michigan. I recently completed my Master's in Computer Science at George Mason University and bring over two years of experience in full-stack and backend development across academic and enterprise environments.

At George Mason University's Contemporary Student Services, I built and deployed web portals using Next.js, Tailwind CSS, and Python/SQL, improving accessibility for over 10,000 students. Previously at Virtusa (Client: Citi), I developed Spring Boot microservices and Kafka-based pipelines handling 100,000+ daily transactions, enhancing system reliability and performance.

This role's focus on coding, application support, and database interaction strongly aligns with my technical background in JavaScript, Python, and SQL and my commitment to secure, high-quality software. I'm particularly drawn to Michigan Engineering's people-first approach and would welcome the opportunity to contribute to its research and educational mission through robust, user-focused applications.

Thank you for your consideration. I look forward to the opportunity to discuss how my skills can support the Robotics Department's initiatives.

Sincerely,

Preethi Ranganathan

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WORK EXPERIENCE

George Mason University – Contemporary Student Services, Fairfax, VA
Web Developer and Data Analyst

Jul 2025 – Present

- **Built and deployed** web applications for Contemporary Student Services using **Next.js** and **Tailwind CSS**, **boosting portal engagement by 30%** and enabling access for **10K+ users**.
- **Developed** backend services and data integration APIs using **Python (Pandas, NumPy)** and **SQL**, and built dynamic visualizations in **Power BI** and **Matplotlib**, enabling trend analysis and improving reporting visibility by **40%** across five student categories for Registrar and Financial Aid teams.

UI Developer

Aug 2023 – May 2025

- **Engineered** front-end redesign of GMU's Patriot Web portal using **React**, **JavaScript**, **MUI**, and **CSS**, **improving navigation and accessibility for 30K+ students**; launched production build on **Railway**.
- **Enhanced** application performance and accessibility by refactoring components and applying modern web standards, resulting in **25% faster page loads** and smoother workflows for students.

George Mason University – Department of Computer Science, Fairfax, VA

Jan 2025 – May 2025

Research Assistant – AI Voice Interaction Automation

- **Engineered** a rooted Android application to simulate mic input in Google Gemini using **OS-level commands** ('am', 'input', 'dumpsys'), **TTS**, and **Accessibility Services**, enabling hands-free voice queries and reducing input delay.
- **Implemented** audio trigger mechanisms with **95% detection accuracy** via real-time **logcat parsing**, validating mic synchronization across **100+ trials**, which minimized latency and removed dependency on ADB.

Virtusa Consulting Services Pvt. Ltd. (Client: Citi – Credit Platform), Hyderabad, India

Software Engineer

Aug 2022 – Aug 2023

- **Developed and released** backend microservices using **Spring Boot**, **Java**, and **REST APIs**, **enhancing reliability for 100K+ daily banking transactions** while **collaborating with distributed Agile teams via Jira across 3 time zones** to ensure timely feature delivery.
- **Integrated Apache Kafka** for real-time credit evaluations, **reducing reconciliation time by 80%** and improving transaction throughput, resulting in faster customer credit decisions.
- **Refactored** legacy **Java/Spring Boot** codebases to **boost API efficiency by 30%** and improve modularity, thereby enabling quicker feature development and easier scaling.
- **Automated** CI/CD pipelines in **Jenkins**, **cutting manual release effort by 40%** and **improving deployment stability**, which led to faster and more consistent production releases.

Associate Engineer

Aug 2021 – Aug 2022

- **Optimized** dashboard batch processing using **PL/SQL**, **improving execution speed by 25%** for **50K+ users**.
- **Developed** internal JSP-based modules with **Spring MVC**, **JavaScript**, and **Oracle DB**, **reducing helpdesk resolution time by 35%**.
- **Executed** functional testing across **10+ Agile sprints**, **identifying 20+ critical bugs** and improving production quality.
- **Managed** UAT and production deployments using **Jenkins** and **Bitbucket**, **improving accuracy and reducing rollbacks**.

EDUCATION

George Mason University, Fairfax, VA

May 2025

Master's in Computer Science, GPA: 3.77

Relevant Coursework: Secure Software Development, UI/UX Design, Mobile Immersive Computing, Data Structures

Jawaharlal Nehru Technological University, Hyderabad, India

Jul 2021

Bachelor of Technology in Information Technology

PROJECTS

Microservices-based Student Survey System

- **Architected** containerized app using **Spring Boot**, **MySQL (Amazon RDS)**, **Docker**, and **Kubernetes(Rancher on AWS EC2)**; automated CI/CD via **GitHub Actions** and **Jenkins**.

Cursor Control via Hand Gestures

- **Designed** a real-time cursor control system using a custom **CNN** trained on hand gesture datasets; integrated **MediaPipe** for landmark detection and gesture recognition, achieving **84% accuracy**.

TECHNICAL SKILLS

Languages/Backend: Java, JavaScript, Python, SQL, PL/SQL, C, C++, Spring Boot, REST APIs, Hibernate

Frontend/UI: React, JSP, MUI, HTML, CSS, Figma, Next.js, Tailwind CSS, Angular, Node.js

DevOps/Tools: AWS (EC2, RDS), Jenkins, Docker, Kubernetes, GitHub Actions, Bitbucket, Jira, Azure, Rancher

Database/Other: Oracle SQL, MySQL, PostgreSQL, JUnit, TDD, AI-assisted coding (GitHub Copilot, Claude, ChatGPT)

Certification: Oracle Certified Professional, Java SE 8 Programmer II